

ROHITH CHALLA

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Education

Pace University, New York, USA

(August 2023 – December 2024)

Master of Science - Data Science- GPA: 3.52

SRM Institute of Science and Technology, Chennai, India

(August 2019 – May 2023)

Bachelor of Technology - Electronic and Computer Engineering - GPA: 8.40

Experience

AI/ML Engineer

(September 2025 – Present)

Perfect Solution groups

- Currently designing, developing, and fine-tuning machine learning and deep learning models, including data preprocessing, algorithm implementation, and real-world integration.
- Writing efficient, well-documented code (Python, TensorFlow, PyTorch, Scikit-learn), testing and optimizing models for accuracy and scalability, and collaborating with teams while staying updated on AI advancements.

Enterprise Automation Associate

(August 2025 – September 2025)

Maturecloud

- Supported the design and development of AI agents and enterprise integrations leveraging AWS, Workday, and Workato cloud platforms.
- Built and configured Workday EIB (Enterprise Interface Builder) integrations to streamline business process workflows and ensure accurate data exchange.
- Developed and deployed Workato automations for enterprise workflows, contributing to scalable, low-code integration solutions across HR and finance processes.

AI Engineer Intern

(February 2025 – August 2025)

Cortracker 360

- Designed and fine-tuned machine learning and deep learning models to solve real-world business problems across multiple domains.
- Fine-tuned Large Language Models (LLMs) like GPT and BERT for domain-specific tasks, improving performance on NLP applications.
- Deployed fine-tuned LLMs into scalable pipelines to deliver efficient, business-aligned AI solutions.

Technical Skills

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| <ul style="list-style-type: none">• Programming languages: Python, SQL, Java• Deep Learning: CNNs, RNNs, LSTMs, GRUs, Transformers, GANs.• Machine Learning Frameworks: TensorFlow, PyTorch, and Scikit-learn.• NLP and Generative AI Models: BERT, Word2Vec, CodeLlama, Google Gemma 2.• Techniques and Applications: Tokenization, Named Entity Recognition (NER), Word Embeddings, Text Summarization, Conversational AI, Retrieval-Augmented Generation (RAG), Sequence-to-Sequence Models, Attention Mechanisms, LLM Fine-tuning, Multi-Agent Systems, and Stateful AI Workflows. | <ul style="list-style-type: none">• Tools and Frameworks: Langchain, Hugging Face Transformers, OpenAI API, Ollama, SQL Toolkit, Agent Type, Crew AI, Lamini, and LangGraph, Whisper, SpeechBrain, Postman, Workato, Workday EIB.• Search and Databases: Hybrid Search with Vector Databases (FAISS, AstraDB), GraphDB, Cypher Query Language, Knowledge Graph-based Q&A.• UI/UX Frameworks: Streamlit and Gradio. |
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Certifications

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| <ul style="list-style-type: none">• AWS Machine Learning Engineer Associate (MLA-C01)• AWS Cloud Practitioner Certificate (CLF-C02)• AWS AI Practitioner (AIF-C01)• Azure AZ-900• Databricks Fundamentals Accreditation | <ul style="list-style-type: none">• SQL Certificate - HackerRank• Hadoop Platform and Application Framework - (UC San Diego)• Deep Neural Networks with PyTorch - (IBM)• Workato Pro Automation I, II, III |
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Projects

AI-Powered Expense Receipt Classifier using Workato and Workday

- Developed a Workato automation that integrated with Workday EIB and RAAS to extract employee expense receipts and process them through OpenAI-powered classification models, automatically distinguishing itemized vs. non-itemized reports.
- Optimized financial workflows by embedding AI-driven validations back into Workday, reducing manual review effort by 70% and improving compliance accuracy across enterprise expense reporting.

Code Assistant – Local LLM-Powered Developer Helper

- Collaboratively built an interactive AI code assistant using a locally hosted LLM (via Ollama) and Gradio, simulating a lightweight pair programmer to deliver real-time code explanations and debugging help.
- Implemented backend APIs with a focus on team extensibility, integrating prompt history and streaming-free inference, reducing latency by 30% and enhancing overall developer productivity.

Speech-to-Speech Emotion Transfer using Whisper and SpeechBrain

- Developed a real-time speech-to-speech emotion transfer system using Whisper and SpeechBrain, achieving 85% emotion accuracy and 90% intelligibility based on 50+ user tests.
- Integrated Whisper (ASR), SpeechBrain (SER + TTS), and Gradio into a lightweight pipeline that runs entirely on CPU, achieving end-to-end inference in under 5 seconds per sample without requiring GPU or external APIs.

Predictive Maintenance Pipeline for Machinery

- Led a cross-functional initiative to build a predictive maintenance system using the NASA turbofan dataset, developing ML models to estimate RUL and reduce unplanned downtime.
- Integrated solutions with AWS S3 and Apache Airflow, improving pipeline scalability and coordination, resulting in a 50% reduction in manual monitoring.